IV. Collagen and Extracellular Matrix

51. Scurvy is a disease of defective collagen synthesis due to deficiency of which vitamin? (Biochemistry Midterm 2025)

Answer: Vitamin C Explanation: Vitamin C is a required co-factor for prolyl and lysyl hydroxylase enzymes that stabilize the collagen triple helix.

52. Vitamin C is essential for the hydroxylation of which amino acids during collagen synthesis? (Biochemistry Midterm 2025)

Answer: Proline and Lysine

53. Which amino acid that is normally part of collagen is not synthesized in Scurvy? (Biochemistry Midterm 2024)

Answer: Hydroxyproline Explanation: Without Vitamin C, the post-translational hydroxylation of proline to hydroxyproline cannot occur.

54. Which of the following amino acids are abundantly found in collagen? (Biochemistry Midterm 2024)

Answer: Glycine and Proline Explanation: Glycine is found at every third position of the polypeptide chain to allow tight packing of the triple helix.

55. Osteogenesis imperfecta results in defective formation of which type of protein? (Biochemistry Midterm 2021)

Answer: Collagen

56. A child with tall stature, loose joints, and detached retinas has a mutation in collagen. Which recurring amino acid is most likely altered? (Biochemistry Midterm 2021)

Answer: Glycine Explanation: Mutations replacing glycine with bulkier amino acids prevent the proper folding of the collagen triple helix.

57. Which copper-dependent enzyme is required for cross-linking in collagen synthesis? (Biochem Proff 2025)

Answer: Lysyl oxidase Explanation: Lysyl oxidase oxidatively deaminates lysine residues to form reactive aldehydes that create stable cross-links.

58. Which post-translational step is most important for the formation of the triple helix of collagen? (Biochem Mid Term Paper Docx)

Answer: Hydroxylation of proline (Hydroxyproline)

59. Which of the following is a non-extracellular matrix protein? (Biochemistry Mids 2020) Answer: Integrin Explanation: Collagen, Elastin, and Laminin are part of the matrix, while Integrins are cell-surface receptors .

60. Soft connective tissues in collagen are often associated with which substance? (Biochem 1st Prof 2016) Answer: Heparin

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V. Immunoglobulins and Specialized Proteins

61. The immunoglobulins with the largest structure is? (Biochemistry Midterm 2021) Answer: IgM Explanation: IgM is a pentamer, meaning it consists of five basic antibody units joined together.

62. Which pentameric immunoglobulin is involved in the primary immune response? (Biochem Mid Term Paper Docx) Answer: Ig-M

63. Which of the following proteins has a length/width ratio of greater than 10? (Biochemistry Midterm 2021) Answer: Fibrinogen

64. Allergies are primarily caused by which type of protein/molecule? (Biochem 1st Prof 2016) Answer: Antibodies (specifically IgE)

65. Which class of protein includes Hemoglobin? (Biochem 2025 Mock DDC) Answer: Conjugated protein Explanation: Hemoglobin is a "heme-protein" because it consists of a globin protein joined to a non-protein prosthetic group (heme) .

66. Which is NOT a classified form of conjugated proteins? (Biochem 2025 Mock DDC) Answer: Incomplete proteins Explanation: Glycoproteins and Lipoproteins are conjugated; "incomplete" refers to dietary nutritional value .

67. Electrophoresis separates proteins mainly on the basis of? (Biochem 2025 Mock DDC) Answer: Charge

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VI. Heme Degradation and Plasma Transport

68. Bilirubin is the end product of heme degradation. How is this hydrophobic molecule made water-soluble? (Biochem Exam 2012) Answer: Conjugation with Glucuronic Acid

69. Unconjugated bilirubin is transported in blood bound to which protein? (Biochem 2025 Mock DDC) Answer: Albumin

70. Heme oxygenase converts heme into which pigment? (Biochem Proff 2025) Answer: Biliverdin

71. Which enzyme is responsible for the initial step of heme degradation? (Biochem Proff 2025) Answer: Heme oxygenase

72. A neonate develops jaundice on day 2 with elevated unconjugated bilirubin. What is the most likely cause? (Biochem 2025 Mock DDC) Answer: Physiological jaundice due to immature UDP-glucuronyl transferase

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VII. Clinical Case-Based Questions (Proteins)

73. A 15-year-old girl with heavy menstrual bleeding and low iron has anemia. What will her peripheral film show? (Biochemistry Midterm 2024) Answer: Microcytic

hypochromic anemia Explanation: Iron deficiency prevents proper heme synthesis, leading to smaller (microcytic) and paler (hypochromic) red blood cells.

74. Kwashiorkor is caused primarily by deficiency of? (Biochem 2025 Mock DDC) Answer: Proteins Explanation: Unlike Marasmus (total calorie deficiency), Kwashiorkor is a selective protein deficiency.

75. Phenylketonuria results from the deficiency of which enzyme? (Biochem 2025 Mock DDC) Answer: Phenylalanine hydroxylase Explanation: This prevents the conversion of phenylalanine to tyrosine.

76. Which organ is most affected by ammonia toxicity? (Biochem 2025 Mock DDC) Answer: Brain Explanation: High ammonia levels interfere with alpha-ketoglutarate in the TCA cycle, depleting energy in neurons.

77. A 3-year-old with hepatosplenomegaly and accumulation of sphingomyelin has which disease? (Biochemistry Midterm 2024) Answer: Niemann-Pick's

78. Which enzyme is defective in Gaucher's disease? (Biochemistry Midterm 2021) Answer: Beta-glucocerebrosidase

79. Increased uric acid level is commonly seen in which condition? (Biochem 2025 Mock DDC) Answer: Gout

80. Deficiency of HGPRT enzyme leads to accumulation of uric acid in which syndrome? (Biochem 2025 Mock DDC) Answer: Lesch-Nyhan syndrome

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VIII. Additional Protein and Nitrogen Metabolism

81. Urea is the primary excretory product of which metabolism in mammals? (Biochem 1st Prof 2016) Answer: Protein/Amino acid metabolism

82. What is the repeating unit of nucleic acids? (Biochem Exam 2012) Answer: Nucleotides

83. A nucleotide is composed of which three components? (Biochem 2025 Mock DDC) Answer: Nitrogenous base + sugar + phosphate

84. Which protein act as a buffer due to their amino and carboxyl terminals? (Biochem Mid Term Paper Docx) Answer: Plasma proteins (and Hemoglobin)

85. The principal buffer of human blood is? (Biochemistry Midterm 2025) Answer: Bicarbonate buffer system Explanation: While proteins buffer, the bicarbonate system is the primary extracellular buffer.

86. Which protein is considered cardio-protective? (Biochemistry Midterm 2021) Answer: HDL (High-Density Lipoprotein)

87. Dietary essential fatty acids (not amino acids) include? (Biochemistry Midterm 2021) Answer: Linoleic acid

88. Which protein transports iron in the blood? (Biochem Exam 2012) Answer: Transferrin

89. What is the storage form of iron in the liver? (Biochemistry Midterm 2024) Answer: Ferritin

90. Lack of intrinsic factor prevents the absorption of which vitamin? (Biochem Exam 2012) Answer: Cobalamin (Vitamin B12)

91. Thalassemia involves the partial or total absence of which chains? (Biochem Exam 2012) Answer: Alpha and/or Beta chains

92. HbA2 is a tetramer composed of? (Biochem Exam 2012) Answer: Two alpha and two delta chains

93. Which amino acid residues are close together to form a disulfide bond? (Biochemistry Mids 2020) Answer: Cysteine

94. What is the function of ferritin regarding iron? (Biochem Exam 2012) Answer: Storage

95. Which type of anemia is characterized by a "pins and needles" sensation and macrocytic MCV? (Biochemistry Midterm 2025) Answer: Vitamin B12 deficiency (Pernicious anemia)

96. Trypsinogen is converted to trypsin by which enzyme? (Biochem Exam 2012) Answer: Enterokinase

97. Which amino acid have a nonpolar side chain? (Biochemistry Midterm 2021) Answer: Phenylalanine

98. The structure of monosaccharides that are mirror images are? (Biochemistry Midterm 2021) Answer: Enantiomers

99. Which nitrogenous base is present only in RNA? (Biochem 2025 Mock DDC) Answer: Uracil

100. Which base is present in DNA but not RNA? (Biochem Exam 2012) Answer: Thymine

101. Translation occurs on which organelle? (Biochem 2025 Mock DDC) Answer: Ribosomes

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